



# DATA SCIENCE

## Weekly Lecture Schedule • Fall 2026

Bachelor of Science in Information Technology • *From Foundations to Intelligent Systems*

Every chapter is seen through four lenses: learning analytics, project management, IoT and cybersecurity

|                                                      |                                                                        |                                                                         |
|------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|
| CONTACT HOURS<br>2 lecture hours + 1 lab hour a week | DURATION<br>16 teaching weeks; midterm Week 8,<br>capstone due Week 16 | CONTENTS<br>32 chapters in 6 parts; Python labs,<br>and R where it fits |
|------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|

| 📅  | 📖 Lecture topics                                                                                                                                                                                                                                                                                                 | 🔧 Laboratory (1 h)                                        | 📝 Assessment             |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------|
| 1  | <b>1 The Data Science Landscape</b> – what data science is, AI-ML-DL nesting, generative and agentic AI, four levels of analytics, team roles<br><b>2 The Data Science Lifecycle</b> – CRISP-DM, framing the problem, two kinds of metric                                                                        | Meet a dataset; write a frame                             | —                        |
| 2  | <b>3 Data: Types, Sources and Quality</b> – three kinds of data, measurement scales, tidy data, quality, provenance<br><b>4 The Mathematical Toolkit</b> – vectors, functions and gradients, gradient descent                                                                                                    | Ten-minute audit; distance and descent                    | —                        |
| 3  | <b>5 Describing Data</b> – centre, spread, shape, common distributions<br><b>6 Probability and Uncertainty</b> – conditional probability, Bayes' theorem, expected value, central limit theorem                                                                                                                  | Centre, spread, shape; base-rate simulation               | <b>Quiz 1</b> (Ch 1–4)   |
| 4  | <b>7 Acquisition, Wrangling and Cleaning</b> – where data comes from, the cleaning sequence, joins, encoding and scaling, data leakage                                                                                                                                                                           | Leak-proof pipeline                                       | —                        |
| 5  | <b>8 Exploratory Data Analysis and Visualisation</b> – choosing the right chart, bivariate exploration, the third variable, visualisation anti-patterns                                                                                                                                                          | Standard EDA pass                                         | <b>Assignment 1 set</b>  |
| 6  | <b>9 Inferential Statistics</b> – confidence intervals, hypothesis testing, effect size and power, multiple comparisons<br><b>10 Relationships and Causality</b> – confounders and colliders, experiments, when you cannot randomise                                                                             | Intervals, tests, confounder simulation                   | —                        |
| 7  | <b>11 Machine Learning Fundamentals</b> – four paradigms, three-way split, overfitting and the learning curve, baselines<br><b>12 Supervised Learning I: Regression</b> – error measures, checking assumptions, polynomial fits, ridge and lasso                                                                 | Split, baseline, diagnose; fit and regularise             | <b>Assignment 1 due</b>  |
| 8  | <b>13 Supervised Learning II: Classification</b> – from regression to classification, four more classifiers, decision boundaries, the threshold as a decision                                                                                                                                                    | Five classifiers compared                                 | <b>Midterm</b> (Ch 1–13) |
| 9  | <b>14 Model Evaluation and Selection</b> – the hinge of Part III: confusion matrix, precision against recall, threshold from costs, cross-validation                                                                                                                                                             | Evaluate honestly; cost-based threshold                   | —                        |
| 10 | <b>15 Feature Engineering and Ensembles</b> – feature selection, ensembles, reading feature importances<br><b>16 Unsupervised Learning and Anomaly Detection</b> – k-means, hierarchical and DBSCAN, dimensionality reduction, anomalies                                                                         | Engineer, ensemble, interpret; cluster and detect         | <b>Assignment 2 set</b>  |
| 11 | <b>17 Neural Network Foundations</b> – one neuron, non-linearity, layers and the forward pass, the training cycle<br><b>18 Deep Architectures</b> – convolutional, recurrent, attention and transformers, autoencoders, transfer learning                                                                        | Network from scratch; transfer learning                   | —                        |
| 12 | <b>19 Sequences, Time Series and Streaming</b> – decomposition, forecasting features, temporal validation, stream processing, drift<br><b>20 Text and Natural Language Processing</b> – bag of words, TF-IDF, dense embeddings, topic modelling                                                                  | Decompose and walk-forward; TF-IDF classifier             | <b>Assignment 2 due</b>  |
| 13 | <b>21 Generative AI and Large Language Models</b> – how an LLM works, adapting a model, prompting, retrieval-augmented generation, evaluation<br><b>22 Agentic AI</b> – agent loop, memory, multi-agent systems, failure modes                                                                                   | Minimal RAG; agent loop with guardrails                   | <b>Quiz 2</b> (Ch 14–22) |
| 14 | <b>23 Data Engineering at Scale</b> – pipelines, where data lives, the five V's, parallelism, data contracts<br><b>24 Cloud and Edge Computing</b> – service models, VMs, containers and serverless, cloud-fog-edge, cost                                                                                        | Data contracts; containerise a model                      | <b>Capstone set</b>      |
| 15 | <b>25 Data Science for the Internet of Things</b> – IoT architecture, telemetry data, sensor fusion, predictive maintenance<br><b>26 Data Science for Cybersecurity</b> – security data sources, alert fatigue, four detection problems, attacking the model                                                     | Telemetry to alert; per-entity baselines and evasion test | —                        |
| 16 | <b>27 MLOps</b> – serving patterns, training-serving skew, versioning, monitoring, safe rollout<br><b>28 Responsible AI and Data Ethics</b> – where bias enters, fairness metrics, privacy, explainability<br><b>29–32 Professional Practice</b> – research methods, project management, communication, capstone | Monitored service; fairness measurement                   | <b>Capstone due</b>      |

### ASSESSMENT WEIGHTS

|                                                  |                        |                                 |
|--------------------------------------------------|------------------------|---------------------------------|
| 20%<br>Sessional: quizzes, assignments, capstone | 30%<br>Midterm, Week 8 | 50%<br>Final-term, all chapters |
|--------------------------------------------------|------------------------|---------------------------------|

**I** Foundations of Data

**IV** Deep Learning and Modern AI

**II** From Data to Insight

**V** Data Science in Systems Context

**III** Machine Learning Core

**VI** Professional Practice

Chapter numbers refer to the Fall 2026 Data Science handbook (Appendix G, Sixteen-Week Teaching Schedule). Chapter 14, Evaluation, is the hinge of the course; Chapters 2, 7, 11, 14 and 28 are relied on by everything downstream. Midterm covers Chapters 1–13; Quiz 2 covers Chapters 14–22; the capstone is marked on the rubric in Chapter 32.